Two Cells, One Closed Box, Merging

Physics, Mechanics, and What Changed from Version 3

BTP — Phase-Field Modelling of Crawling Cells

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Abstract

This note documents the physics and the control logic implemented in `BTP_version3_two_cell_walk.ipynb`: two independently crawling cells, each repeatedly choosing a direction, deforming to move that way, relaxing back towards a circle, and choosing again, confined to a closed box, that coalesce into one cell when their paths cross. The underlying continuum model is exactly Version 3's full phase-field / active-polar-fluid description with a spectral Stokes solve; this note is organised around *what changed* to produce the new behaviour, and why each change was made the way it was. **The notebook has not been executed as of this writing**; Section 5 describes the designed, expected behaviour and the diagnostics the code will produce, not measured output, and says so explicitly wherever a number would otherwise look like a result.

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1 Scope and an upfront caveat

`BTP_version3_two_cell_walk.ipynb` was written and statically checked (every cell compiles; a `pyflakes` pass returns no warnings) but **deliberately not run**, per the project owner's request to execute new code on their own machine. Consequently this note cannot report measured speeds, merge times, or rendered figures. What it *can* do, and what it does below, is explain precisely what the model computes, why each new mechanism was added, how it differs from Version 3 and from the earlier Version 4 two-cell experiment, and what the diagnostics are designed to show once the notebook is run. Every quantitative parameter below (box size, wall strength, relaxation time, distance ranges) is a *chosen input*, not a fitted or verified one.

2 Recap: the Version 3 physics, unchanged

The two-cell notebook keeps the entire single-cell Version 3 model intact; nothing in this section was modified. It is restated here because every later section refers back to these equations by name.

2.1 Fields

A cell is a diffuse phase field $\phi(x, y, t)$, equal to 1 inside the cell and 0 outside, with a smooth interface of width set by ε. The actin cytoskeleton's local orientation and polymerisation activity is a vector field $\mathbf{P}(x, y, t)$, non-zero only where $\phi > 0$. The cytoplasm's velocity is $\mathbf{v}(x, y, t)$, obtained at every instant from a force balance (not integrated forward in time — see Section 3.7).

2.2 Free energy

Following Tjhung, Tiribocchi, Marenduzzo and Cates [1],

$$F[\phi, \mathbf{P}] = \int d^2r \Big\{ \underbrace{a\,\phi^2(\phi-1)^2}_{\text{double well}} + \underbrace{\tfrac{\varepsilon^2}{2}|\nabla\phi|^2}_{\text{interfacial tension}} - \underbrace{\tfrac{\alpha}{2}(2\phi-1)|\mathbf{P}|^2 + \tfrac{\alpha}{4}|\mathbf{P}|^4}_{\text{polar order inside the cell}} + \underbrace{\tfrac{\kappa}{2}(\nabla\mathbf{P})^2}_{\text{Frank elasticity}} + \underbrace{\beta\,\mathbf{P}\cdot\nabla\phi}_{\text{anchoring}} \Big\}. \tag{1}$$

The double well plus the interfacial term give the cell a boundary with a cost; the α terms drive $|\mathbf{P}|^2 \to 2\phi - 1$ strictly inside the cell; the Frank term penalises sharp turns in $\mathbf{P}$; and the anchoring term, with $\beta > 0$, makes $\mathbf{P}$ prefer to point outward, normal to the local boundary — in every version of this project, β has been the dominant shape parameter.

Differentiating gives the chemical potential driving ϕ and the molecular field driving $\mathbf{P}$:

$$\mu_\phi = \frac{\delta F}{\delta \phi} = 2a\,\phi(\phi-1)(2\phi-1) - \varepsilon^2\nabla^2\phi - \alpha|\mathbf{P}|^2 - \beta\,\nabla\cdot\mathbf{P}, \tag{2}$$

$$\mathbf{h} = \frac{\delta F}{\delta \mathbf{P}} = \big[-\alpha(2\phi-1) + \alpha|\mathbf{P}|^2\big]\mathbf{P} - \kappa\nabla^2\mathbf{P} + \beta\nabla\phi. \tag{3}$$

2.3 Equations of motion

With $\mathbf{u} = \mathbf{v} + w_0\mathbf{P}$ the net velocity transporting cell material (w_0 the actin treadmilling rate),

$$\partial_t\phi + \nabla\cdot(\phi\,\mathbf{u}) = -M\big[\mu_\phi - \Lambda\,g(\phi)\big], \qquad g(\phi) = \phi(1-\phi), \tag{4}$$

$$\partial_t\mathbf{P} + (\mathbf{u}\cdot\nabla)\mathbf{P} = -\boldsymbol{\Omega}\cdot\mathbf{P} + \chi\,\mathbf{D}\cdot\mathbf{P} - \mathbf{h}/\Gamma, \tag{5}$$

where $\mathbf{D}$ and $\boldsymbol{\Omega}$ are the symmetric (strain-rate) and antisymmetric (vorticity) parts of $\partial_\alpha v_\beta$, Γ a rotational viscosity, and χ the flow-alignment parameter. Λ is a spatially uniform Lagrange

multiplier, $\Lambda = \int \mu_\phi \, dA / \int g \, dA$, which makes Eq. (4) conserve $\int \phi \, dA$ *exactly*, to machine precision, by construction — this is the Rubinstein–Sternberg conservative form of Allen–Cahn dynamics [3], not an approximate correction.

2.4 Force balance

At the cell's scale $Re \sim 10^{-5}$, so inertia is dropped and the force balance is the Stokes limit,

$$0 = -\nabla p + \eta \nabla^2 \mathbf{v} - \xi \mathbf{v} + \mathbf{f}, \qquad \nabla \cdot \mathbf{v} = 0, \tag{6}$$

with the substrate contribution $-\xi \mathbf{v}$ standing in for the wall of the original 3D geometry (the continuum limit of a carpet of stochastic adhesive linkers [2]), and the body force

$$f_\alpha = \partial_\beta \big(\sigma^{\mathrm{act}}_{\alpha\beta} + \sigma^{\mathrm{el}}_{\alpha\beta} \big) - \phi \, \partial_\alpha \mu_\phi, \qquad \sigma^{\mathrm{act}}_{\alpha\beta} = -\zeta \, \phi \, P_\alpha P_\beta, \tag{7}$$

$\zeta < 0$ contractile, σ^{el} the elastic (Frank + co-rotational) stress, and the last term of Eq. (7) the capillary force in potential form.

2.5 The spectral solve

Because the domain is periodic, Eq. (6) is solved exactly in Fourier space: a Leray (transverse) projection eliminates the pressure, and dividing by $\xi + \eta k^2$ gives $\hat{\mathbf{v}}(\mathbf{k})$ directly — one pair of FFTs per step, no iteration, no time-step restriction from the viscous term. **This spectral method is the one piece of Version 3's numerics that only works because the domain is periodic**, a fact Section 3.2 returns to.

3 What changed: from one cell to two, in a closed box

Table 1 gives the short version; the rest of this section explains each row.

	Version 3 (single cell)	**Two-cell walk notebook**
Grid	N=128, L=80	N=192, L=120 (room for two cells and a walk)
Domain	periodic, infinite in effect	periodic numerically, but confined by a new soft wall term
Phase/polarization fields	one ϕ, one $\mathbf{P}$, one cell	one ϕ, one $\mathbf{P}$, *two disjoint blobs* at t=0
Activity (w_0, ζ, β)	uniform scalars, on for the whole run	w_0 becomes a *field* (array); each blob's region is independently on/off
Direction of $\mathbf{P}$	set once at t=0, implicit in the initial condition	chosen explicitly, re-chosen on a schedule, per blob
What stops the cell at the edge	nothing (recentring trick keeps one cell near the middle)	the new wall potential (no recentring; absolute position matters)
Two cells touching	not modelled	coalesce into one connected ϕ region, detected via connected-component labelling
Post-collision direction	n/a	biased towards the two cells' combined pre-collision momentum

Table 1: What changed, at a glance.

3.1 Grid and domain

The grid grew from 128×128 ($L = 80$) to 192×192 ($L = 120$) at the same resolution $dx = 0.625$. Two cells need room to start apart, walk, and potentially collide without ever nearing the box edge closely enough for the soft wall (Section 3.2) to distort their shape before a genuine wall-avoidance decision should fire.

3.2 The soft confining wall

Version 3's Stokes solve is spectral, which requires periodicity; there is no way to impose a literal no-slip or free-slip wall inside that method without replacing it with a different linear solve (finite differences or finite volumes, discretising Eq. (6) with wall boundary conditions and solving the resulting sparse system — a substantial rewrite, not attempted here). Instead, exactly *one* new term is added to the chemical potential of Eq. (2):

$$\mu_\phi \;\longrightarrow\; \mu_\phi + k_{\text{wall}}\, W(x,y)\, \phi, \qquad W(x,y) = \left[\frac{\max\big(0,\, |x| - x_0\big)^2 + \max\big(0,\, |y| - y_0\big)^2}{(L/2 - x_0)^2} \right]^{1/2 \cdot \text{sharp}}, \tag{8}$$

precomputed once as a static array W: zero everywhere inside a square region starting at $x_0 = y_0 = 0.8 \times (L/2)$ from the centre, rising smoothly (quartic by default, set by `wall_sharpness`) to 1 at the literal domain edge. This is structurally the *same kind of term* as the pairwise steric-repulsion used in the Version 4 two-cell experiment ($+2k_{\text{rep}}\phi_i\phi_j^2$ in each cell's μ_i) — a quartic penalty that only activates where ϕ overlaps a forbidden region — except the forbidden region here is a fixed wall field W instead of another cell's ϕ_j. It plugs into the unmodified relaxation term of Eq. (4) with no other code change.

What this is not. It is a soft potential, not a boundary condition. The fluid velocity field $\mathbf{v}$ is still computed on a periodic domain; what changes is that ϕ is discouraged, by an energy penalty, from occupying the near-edge zone, which behaviourally keeps the cell inside a confined region without literally terminating the computational domain there. The strength $k_{\text{wall}} = 40$ and the margin $0.8 \times (L/2)$ are *reasoned starting values*: they were chosen by the same reasoning used for the Version 4 repulsion coefficient (large enough, relative to the double-well coefficient a, to resist the active crawling force pushing against it) but were not empirically tuned by running this notebook. A runtime check (`touches_boundary`) prints a warning if $\phi > 0.5$ ever reaches within six grid cells of the literal array edge, which would indicate k_{wall} needs to be raised.

3.3 Two blobs, one field

The Version 4 two-cell experiment (a separate, earlier notebook) used *two separate* fields, $\phi_1, \mathbf{P}_1$ and $\phi_2, \mathbf{P}_2$, coupled by a steric term specifically so the cells would stay distinguishable and never overlap. This notebook does the opposite on purpose: it uses *one* ϕ and *one* $\mathbf{P}$, initialised with two disjoint tanh profiles,

$$\phi(\mathbf{r}, 0) = \max\left(\tfrac{1}{2}\big[1 - \tanh(\tfrac{|\mathbf{r}-\mathbf{r}_1| - R_0}{\sqrt{2}\,\varepsilon})\big],\; \tfrac{1}{2}\big[1 - \tanh(\tfrac{|\mathbf{r}-\mathbf{r}_2| - R_0}{\sqrt{2}\,\varepsilon})\big] \right), \tag{9}$$

and $\mathbf{P}$ set independently within each blob's own disc to point along that blob's chosen heading. While the two regions of $\phi > 0$ are disjoint, nothing distinguishes them as separate objects to the underlying PDE — Eqs. (4)–(5) are evaluated over the whole grid every step, exactly as they are for one cell. When the two regions' boundaries touch, the double-well and interfacial-tension

terms of Eq. (1) do what they always do: minimise interfacial length, which for two adjacent blobs of the *same* field means rounding the pair into one connected shape. **No merging algorithm was written**; coalescence is a direct consequence of using one shared field, which is the most literal reading of "mix into one another" available in a diffuse-interface model — once the interface between them disappears, there is no record left of which material came from which original cell.

3.4 Activity as a field, not a scalar

Before the two blobs merge, one of them may be actively crawling (treadmilling, anchoring, active stress all switched on) while the other is relaxing (all switched off), simultaneously, in the same field. This requires only one structural change: w_0, previously a single scalar multiplying $\mathbf{P}$ in $\mathbf{u} = \mathbf{v} + w_0\mathbf{P}$, becomes a spatial array $w_0(x, y)$, equal to the active value inside whichever blob is currently in its "move" phase and zero elsewhere (including inside a blob that is relaxing). ζ and β needed no such change: they already appear multiplying other fields elementwise in Eqs. (2)–(7) (`chemical_potential_phi`, `molecular_field_P`, `body_force`), so passing them in as arrays instead of scalars works with the existing code unmodified — only the function signature and the one line computing $\mathbf{u}$ needed to accept a field for w_0.

3.5 The decision layer

Each disjoint blob (and, after merging, the single resulting cell) carries a small state: a current heading θ, a phase (*move* or *relax*), a target travel distance for the current move leg, and the position where that leg began. Every `sample_every` steps (= 1 time unit at $dt = 0.01$), three conditions are checked, in priority order:

1. **Near a wall.** If the blob's centroid is within `wall_redirect_margin` of the box edge, pick a new heading biased back toward the box centre (a von Mises draw centred on $\arctan 2(-y, -x)$, concentration `wall_kappa`) and start a new move leg immediately.

2. **Move leg finished.** If the blob is in *move* phase and has covered its target distance, switch to *relax*: w_0, ζ, β all set to zero within its mask. With no energy sustaining the fan shape, Eq. (4)'s relaxation term — the same interfacial relaxation validated in Version 3's passive-limit test, an ellipse relaxing to a circle — pulls the blob back towards a circle on its own; nothing new was added for this.

3. **Relax finished.** If the blob has been relaxing for at least `relax_time`, pick a fresh, uniformly random heading and resume *move*.

Identifying which blob is which between checks uses `scipy.ndimage.label` on $\phi > 0.5$ to find connected components, matched to the previous check's two tracked blobs by whichever pairing minimises total centroid displacement — this keeps a blob's identity (and hence its in-progress target distance and heading) stable across checks even though the labelling algorithm itself has no memory of which numbered label is which cell.

3.6 The merge event

The moment `ndimage.label` reports **one** connected component where it previously reported two, the merge branch fires exactly once. Using the *last* check before contact (where the two blobs were still separately resolvable), it computes each blob's mass-like weight $m_i = \int_{\Omega_i} \phi \, dA$, its instantaneous crawling velocity $\mathbf{v}_i$ (centroid displacement over the preceding check interval, divided by the

elapsed time), and from these:

$$\mathbf{p}_i = m_i\,\mathbf{v}_i, \qquad \mathbf{p}_{\text{tot}} = \mathbf{p}_1 + \mathbf{p}_2, \tag{10}$$

$$\mathbf{R}_{\text{cm}} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}, \qquad L_i = m_i\,(\mathbf{r}_i - \mathbf{R}_{\text{cm}}) \times \mathbf{v}_i, \qquad L_{\text{tot}} = L_1 + L_2. \tag{11}$$

L_i is the z-component of the orbital angular momentum each blob's crawling motion carries about the eventual contact point — exactly the quantity this project's adviser asked to be taken into account. Section 3.7 explains why it is used as a *bias*, via a von Mises distribution centred on $\arctan 2(p_{\text{tot},y}, p_{\text{tot},x})$ with concentration `merge_kappa`, for the merged cell's first new heading — rather than enforced as a literal conserved quantity. $\mathbf{p}_{\text{tot}}$ and L_{tot} are both printed when the merge fires, as a diagnostic record of what the pre-collision state implied, whether or not the model's own stochastic choice ends up close to that direction.

3.7 Why not literal angular-momentum conservation

This is worth stating plainly rather than quietly working around. Stokes flow (Eq. (6)) is the zero-Reynolds-number limit: inertia has been dropped from the equations entirely, which is also why this project has used $Re \sim 10^{-5}$ to justify the force balance since Version 3 was first built. In that limit, momentum is *not* a quantity carried forward in time at all — net force and net torque on the fluid are balanced *instantaneously*, every step, by viscous dissipation and the substrate friction $-\xi\mathbf{v}$, which is itself an external drag that can (and does) absorb momentum from outside the fluid/cell system. There is no dynamical bookkeeping in Eq. (6) analogous to $\frac{d}{dt}(m\mathbf{v}) = 0$ for an isolated collision, because there is no inertial term to begin with. This is the standard, if counter-intuitive, fact about swimming at low Reynolds number: a cell that stops generating active stress stops moving *immediately*, it does not coast [4].

The notebook therefore does not — and physically should not — enforce $\mathbf{p}_{\text{tot}}$ or L_{tot} as a literal post-collision constraint on the merged cell's velocity or rotation; nothing in Eq. (6) would support such a constraint without breaking the force-balance structure the whole model rests on. What it does instead, as described in Section 3.6, is treat $\mathbf{p}_{\text{tot}}$ as *information*, biasing a stochastic decision the model was always going to make (a new random heading) rather than determining it outright. This keeps faith with the physics while still giving the collision's approach geometry a say in what happens next, which is the practical substance of the adviser's suggestion.

4 Mechanics of the control loop

Algorithmically, one iteration of the outer loop (one `sample_every` interval, ≈ 1 time unit):

1. Label $\phi > 0.5$ into connected components.
2. *If not yet merged and two components are found:* identify and match them to the two tracked states (Section 3.5); for each, evaluate the three triggers in order and, if one fires, update that blob's state and repoint $\mathbf{P}$ within its mask (`set_direction`); build the elementwise activity fields $\zeta_{\text{run}}, \beta_{\text{run}}, w_0$ from both blobs' current phases.
3. *If not yet merged and one component is found:* run the merge computation of Eq. (11), collapse to a single tracked state with a momentum-biased new heading, repoint $\mathbf{P}$ over the whole merged region, and report $\mathbf{p}_{\text{tot}}, L_{\text{tot}}$.
4. *If already merged:* evaluate the same three triggers for the single state.

5. Advance `sample_every` individual timesteps of Eqs. (4)–(5) and (6) (`advance_one_step`), unchanged from Version 3 except that w_0 is now an array; store a video frame every `frame_every` steps, and check for divergence.

The loop ends when the elapsed simulated time reaches `t_final`.

5 Diagnostics and expected behaviour

No numbers in this section were measured — the notebook has not been run. This section describes what each diagnostic is designed to show, and the qualitative behaviour the design is intended to produce, so that a reader can judge the first real run against it.

Path plot Each tracked entity's centroid is logged every check interval, tagged by identity (`0`, `1`, or `merged`) and phase (*move* or *relax*). The intended picture: two separate, roughly piecewise-linear trajectories (straight during *move*, nearly stationary during *relax*) that converge, mark a merge point, and continue as one trajectory thereafter — visually distinguishing "two independent walks" from "one combined walk" without needing to inspect the field directly.

Merge report $\mathbf{p}_{\text{tot}}$, L_{tot}, and the resulting new heading, printed once. If the two cells approach nearly head-on with similar masses, $\mathbf{p}_{\text{tot}}$ should be small and the new heading close to uniformly random (the von Mises bias has little to act on); if they approach at an angle with different masses or speeds, $\mathbf{p}_{\text{tot}}$ should point roughly along the net approach direction and bias the merged cell's first leg that way.

Video `two_cell_walk.mp4`, titled per frame with each active entity's phase and (while moving) heading in degrees, intended to make the whole decide/move/relax/redirect cycle — and the moment of coalescence — legible without reading the printed log alongside it.

Boundary warning `touches_boundary` should *not* fire during a normal run; if it does, k_{wall} or the wall margin need to be increased (Section 3.2).

Expected qualitative sequence. Two blobs, placed `init_separation` $= 50$ apart along x, each independently cycle through move/relax legs of random length drawn from $(15, 35)$ and random heading, inside a box of half-width 60 with the wall zone beginning at 48. Given random, independent headings, whether and when the two paths cross is itself random — for a `t_final`$= 400$ run there is no guarantee a merge event occurs at all; the run should be repeated with a different `seed` or a larger `t_final`/smaller `init_separation` if it does not. The notebook prints an explicit message distinguishing a run with a merge from one without, rather than assuming one will happen.

6 Open questions and known limitations

- **The wall is a potential, not a boundary condition** (Section 3.2); k_{wall} and the margin are untuned.
- **Relaxation uses a fixed duration**, not the circularity-based criterion of the single-cell run-and-tumble notebook — simpler, at the cost of not actually checking the blob has rounded off before resuming activity.

- **The merge biases heading only.** A natural extension, not implemented, is seeding a small swirl in $\mathbf{P}$ proportional to the sign of L_{tot}, so the merged cell visibly starts to curve rather than only picking a biased straight heading.
- **Connected-component labelling can be confused by small numerical artefacts.** The code defends against spurious extra components by always tracking only the two largest, but has not been stress-tested against this failure mode by execution.
- **Generalising past two cells** needs the `states` list and the nearest-centroid matching logic generalised from 2 to N blobs; not attempted here.
- **The honest long-term fix for Section 3.2** is a genuine non-periodic Stokes solve (a finite-difference Poisson/Helmholtz solve with wall boundary conditions, replacing the two-FFT spectral method with a sparse linear solve) rather than a soft potential on a periodic domain.

References

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